

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I C. Sai Harshitha (192424346) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

C. Sai Harshitha

Annexure A

Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

A hospital can develop an intelligent diagnosis system using **Bayes' Theorem**, **Naïve Bayes**, and **Bayesian Belief Networks** to predict diseases from patient symptoms and medical history. These methods help handle uncertainty and update predictions when new information is available.

1. Bayes' Theorem

Bayes' Theorem calculates the probability of a disease based on observed symptoms.

$$P(D|S) = [P(S|D) \times P(D)] / P(S)$$

Where $P(D|S)$ is the probability of disease given the symptoms.

Fever, cough, and fatigue can be used to estimate the probability of influenza or pneumonia.

2. Naïve Bayes Classifier

Naïve Bayes classifies the patient into the disease with the highest probability. It assumes that symptoms are conditionally independent given the disease.

$$P(D|S_1, S_2, \dots, S_n) \propto P(D) \times P(S_1|D) \times P(S_2|D) \times \dots \times P(S_n|D)$$

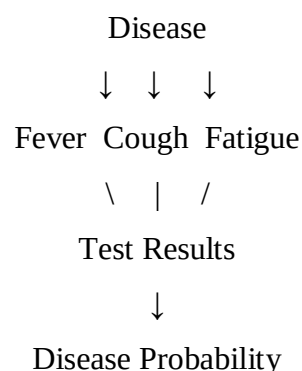
Example:

Disease	Probability
Influenza	0.75
Pneumonia	0.18
Other	0.07

Therefore, the system predicts **Influenza**.

3. Bayesian Belief Network

A BBN represents relationships between diseases, symptoms, and test results.



It helps represent **conditional dependencies** and update the disease probability when new evidence is available.

4. Handling Uncertainty

Medical diagnosis is uncertain because the same symptom can occur in different diseases and some information may be missing or inaccurate.

The system represents this uncertainty using probabilities.

Example: Probability of Influenza = 75%

5. Probability Learning

The system learns probabilities from previous patient records, such as:

- Disease occurrence
- Symptoms associated with diseases
- Medical test results
- Relationships between symptoms and diseases

These probabilities can be updated as new patient data becomes available.

6. Effectiveness and Constraints

The approach can handle incomplete information, combine multiple symptoms, and provide probability-based predictions. However, its effectiveness depends on **data quality and sufficient training data**.

Important constraints include:

- Patient data privacy
- Missing or incorrect records
- Dependencies between symptoms
- Need for clinical validation

Conclusion

Bayes' Theorem provides the probability calculation, **Naïve Bayes performs disease classification**, and **Bayesian Belief Networks represent conditional dependencies**. Probability learning allows the system to improve from patient data, making it useful for supporting medical diagnosis while considering real-world limitations.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

An organization can develop a **spam detection system** using **Naïve Bayes and Bayes Optimal Classifier** to classify emails as spam or not spam. The system can also use **Maximum Likelihood (ML)** and **Minimum Description Length (MDL)** for feature selection.

1. Naïve Bayes for Spam Detection

Naïve Bayes calculates the probability that an email belongs to each class based on its features such as **words, links, and attachments**.

$$P(\text{Spam}|\text{Features}) \propto P(\text{Spam}) \times P(\text{Features}|\text{Spam})$$

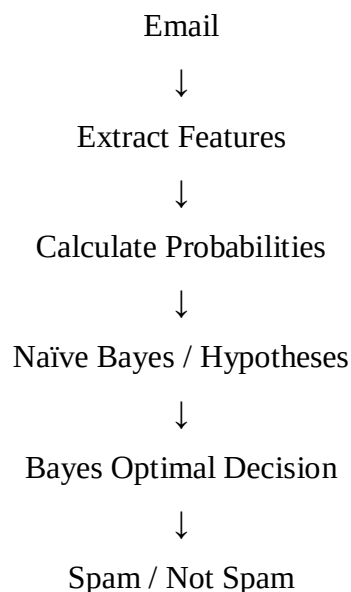
Example:

Class	Probability
Spam	0.85
Not Spam	0.15

Since spam has the higher probability, the email is classified as **Spam**.

2. Bayes Optimal Classifier

The Bayes Optimal Classifier considers the predictions of all possible hypotheses and selects the class with the highest overall posterior probability.



It can provide better decisions when there is uncertainty between different possible models.

3. Maximum Likelihood vs MDL

Maximum Likelihood (ML):

- Selects parameters or features that make the observed training data most likely.

- Focuses mainly on fitting the available data.
- Can lead to complex models when too many features are used.

Minimum Description Length (MDL):

- Selects a model that gives the shortest overall description of the **model and data**.
- Balances model complexity and accuracy.
- Helps reduce unnecessary features and overfitting.

ML	MDL
Focuses on data likelihood	Focuses on data + model complexity
May produce complex models	Prefers simpler models
Can overfit	Helps reduce overfitting

4. Performance and Trade-offs

The spam detector can be evaluated using **accuracy, precision, recall, and error rate**.

A complex model may capture more patterns but require more computation and may overfit. A simpler model is faster and easier to maintain but may miss some important patterns.

5. Effect of Sample Size and Hypothesis Space

Sample size:

More training emails generally provide better estimates of probabilities and improve classification reliability. A small dataset may produce inaccurate probability estimates.

Hypothesis space:

A larger hypothesis space provides more possible models but requires more data to identify the appropriate model. A smaller hypothesis space is easier to learn but may not capture all spam patterns.

Conclusion

A spam detection system can combine **Naïve Bayes and Bayes Optimal Classification** to make probabilistic email classifications. **Maximum Likelihood** focuses on fitting the observed data, while **MDL** considers both accuracy and model complexity. Increasing the sample size generally improves learning, while the size of the hypothesis space affects model complexity and the amount of data required.

3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.

A retail company can use the **Expectation-Maximization (EM) algorithm** to identify hidden customer groups from purchasing data, even when some information is incomplete. The algorithm estimates the underlying probability distributions and assigns customers to suitable clusters.

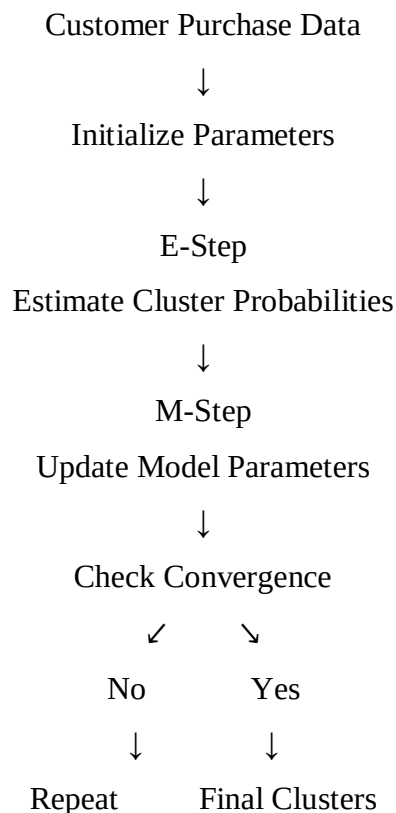
1. EM Algorithm: EM works in two main steps:

E-Step (Expectation):

Estimate the probability that each customer belongs to each cluster using the current model parameters.

M-Step (Maximization):

Update the model parameters using the probabilities calculated in the E-Step.



2. Customer Clusters

For example, customers may be grouped as:

Cluster	Customer Behaviour
C1	Frequent, high-value purchases
C2	Occasional purchases
C3	Low-value purchases

The probabilities obtained by EM help determine the most suitable cluster for each customer.

3. Convergence and Initialization

EM starts with initial parameter values and repeatedly performs the E-Step and M-Step until the results become stable.

Different initial values can produce different clusters. Therefore, **good initialization** is important, and multiple initializations can be used to obtain better results.

4. Probability Learning

EM learns the underlying probability distributions from the available purchasing data. It is useful when some customer information is **hidden or incomplete**, because it estimates the missing information during the learning process.

5. Use in Targeted Marketing

The identified clusters can help the company create targeted marketing strategies:

- **High-value customers:** premium offers and loyalty rewards
- **Occasional customers:** discounts to encourage frequent purchases
- **Low-value customers:** personalized promotions

6. Limitations

The main limitations are:

- Sensitive to initialization
- Can converge to a local optimum
- Requires suitable probability assumptions
- Incomplete or poor-quality data can affect results
- Selecting the appropriate number of clusters can be difficult

Conclusion

The EM algorithm can effectively segment customers by estimating hidden patterns and probability distributions from purchasing data. Its iterative E-Step and M-Step help identify meaningful customer groups, which can be used for targeted marketing. However, initialization, convergence, data quality, and model assumptions must be considered.

4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labelled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.

A research team can develop a **concept learning system** to classify medical images as **normal or abnormal**. The system can use **hypothesis space search and Candidate Elimination** to learn from labelled examples and generalize to unseen images.

1. Concept Learning

Concept learning learns a general rule from labelled training examples.

For this problem:

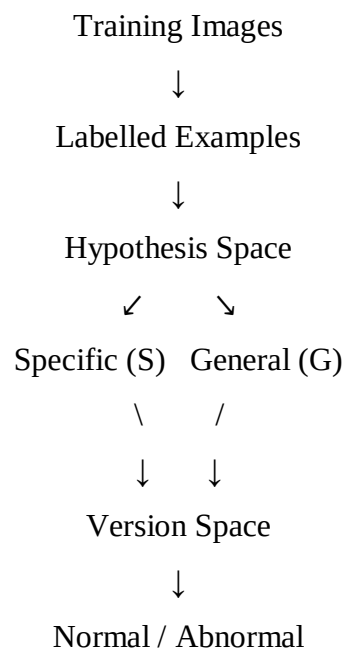
- **Positive examples:** Abnormal medical images
- **Negative examples:** Normal medical images

The system searches the **hypothesis space** to find rules that correctly classify the training images.

2. Candidate Elimination

Candidate Elimination maintains two boundaries:

- **S – Specific boundary:** Most specific hypotheses that fit the examples.
- **G – General boundary:** Most general hypotheses that fit the examples.



When a new labelled example is given, the **S and G boundaries are updated** to remove inconsistent hypotheses.

3. Finite vs Infinite Hypothesis Spaces

Finite hypothesis space:

Contains a limited number of possible hypotheses. It is easier to search and requires fewer computational resources.

Infinite hypothesis space:

Contains an unlimited number of possible hypotheses. It provides greater flexibility but makes learning and finding the correct hypothesis more difficult.

4. Sample Complexity

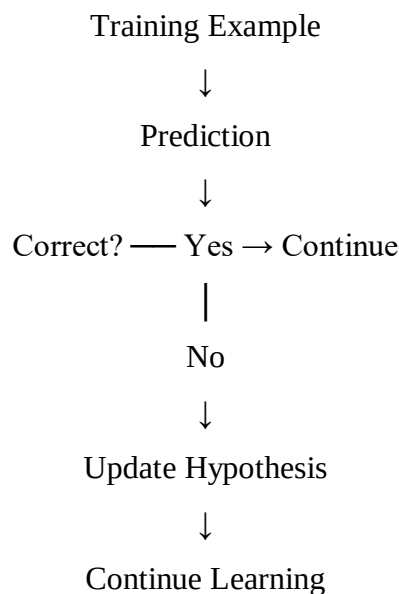
Sample complexity refers to the **amount of labelled training data required** for a learning system to achieve reliable performance.

For medical images, sufficient examples of both normal and abnormal cases are required. More representative training examples generally help the system learn better and generalize to new images.

5. Mistake Bound Model

The mistake bound model measures learning performance by considering the **number of incorrect predictions made before the learner identifies a suitable hypothesis**.

A lower number of mistakes indicates better learning performance.



6. Generalization to Unseen Data

After learning from labelled medical images, the system should correctly classify new images that were not included in training.

Good generalization depends on:

- Sufficient training examples
- Representative medical images
- Suitable hypothesis space
- Correctly labelled data
- Avoiding overfitting

5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

1. Bayesian Inference

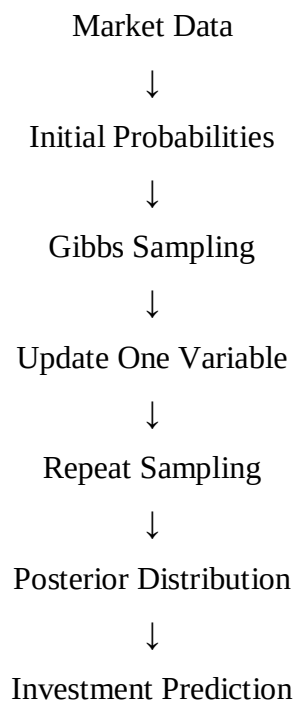
Bayesian inference combines prior knowledge with new market evidence to calculate the updated probability of an investment outcome.

$$\text{Posterior Probability} \propto \text{Likelihood} \times \text{Prior Probability}$$

For example, the system can estimate the probability of an investment giving a **high, medium, or low return** based on previous market trends and current information.

2. Gibbs Algorithm

The Gibbs algorithm can be used to estimate unknown variables in a probabilistic model by repeatedly sampling one variable while keeping the others fixed.



Repeated sampling helps the system estimate the underlying probability distribution.

3. Role of Prior Knowledge

Prior knowledge represents what is already known about the market before observing new data.

For example:

Prior belief → New Market Data → Updated Belief

If previous market data suggests a high probability of positive returns, this prior belief can influence the initial prediction. New evidence can then increase or decrease the probability.

4. Risk Evaluation Using Posterior Probability

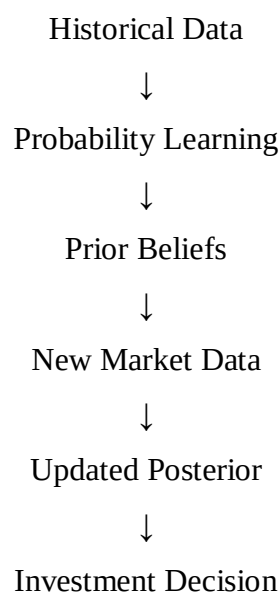
Posterior probabilities can be used to evaluate investment risk.

Investment Outcome	Posterior Probability
High Return	0.60
Moderate Return	0.25
Low Return	0.15

The institution can use these probabilities to compare investment options and make better decisions under uncertainty.

5. Probability Learning

The system learns probability distributions from historical market data. As new information becomes available, the model updates its probabilities.



6. Computational Challenges and Optimization

Bayesian models and Gibbs sampling may require many iterations, making them computationally expensive for large datasets.

The system can improve efficiency by:

- Using suitable initialization
- Reducing unnecessary iterations
- Using efficient sampling methods
- Processing relevant market features

Conclusion

Bayesian inference combines **prior knowledge and new evidence** to update investment predictions. The **Gibbs algorithm** estimates posterior distributions through repeated sampling, while probability learning allows the system to adapt as new market data becomes available. Posterior probabilities can then be used to evaluate investment risk and support better decisions under uncertain market conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand